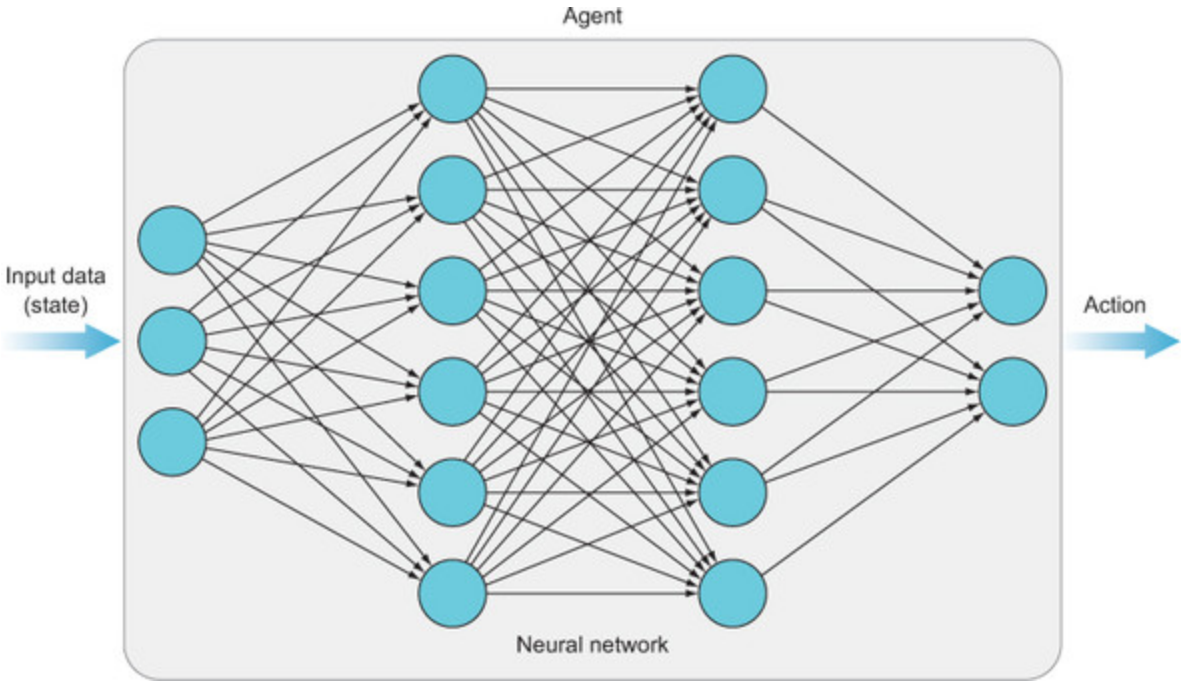
*network*s, see figure 1.9). But remember, RL is more about the type of problem and solution than about any particular learning algorithm, and you could certainly use alternatives to deep neural networks. In fact, in chapter 3 we'll begin by using a very simple non-neural network algorithm, and we'll replace it with a neural network by the end of the chapter.

Figure 1.9. The input data (which is the state of the environment at some point in time) is fed into the agent (implemented as a deep neural network in this book), which then evaluates that data in order to take an action. The process is a little more involved than shown here, but this captures the essence.



The agent's only objective is to maximize its expected rewards in the long term. It just repeats this cycle: process the state information, decide what action to take, see if it gets a reward, observe the new state, take another action, and so on. If we set all this up correctly, the agent will eventually learn to understand its environment and make reliably good decisions at every step. This general mechanism can be applied to autonomous vehicles, chatbots, robotics, automated stock trading, healthcare, and much more. We'll explore some of these applications in the next section and throughout this book.